

RUOHAN LIU

Tidmarsh Lane 14 | Oxford OX1 1AZ
+44 07856124342 | ruohan.liu@ox.ac.uk

EDUCATION

OXFORD UNIVERSITY

Master of Research MRes Clinical Neurosciences// Supervised by Pr Ben Seymour & Tim Denison

Oxford-UK
2023-Present

- Under the division of Nuffield Clinical Neurosciences (NDCN) and Institute of Biomedical engineering (IBME)
- Project design using neuroimaging studies on the prevention of sport injury through examination of mental toughness and 'effort-reward paradox'

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

Visiting student in Department of Brain and Cognitive sciences// Supervised by Pr James di Carlo

Boston, US
Present

- Studying object recognition in macaques using deep brain stimulation in the inferior temporal cortex

UNIVERSITY COLLEGE LONDON

Bachelor of Science With Honours, BSC Biomedical Science // Supervised by Pr Patrick Haggard

London-UK
2020 - 2023

- Concentration: Neuroscience (Computational and Cognition)
- Relevant courses: Pharmacology, Neural Computation, System Neuroscience, Cognitive Neuroscience, Human Neuroanatomy, Biochemistry, Biophysics

GATSBY CENTRE OF COMPUTATIONAL NEUROSCIENCES

Summer Bridging Programme on Maths for Theoretical neurosciences

London -UK
2023

- Full time summer course exploring three general math's topics: Linear Algebra, Calculus, Probability
- Coding workshops and tutorials to apply this knowledge into context of machine learning, neural networks, etc

BRITISH COUNCIL SCHOOL OF MADRID

Spanish Baccalaureate (Evau) 12.9/14

Madrid, Spain
2014 - 2020

Highschool

- Relevant classes: Physics, Chemistry, Biology, Maths, Psychology

RESEARCH EXPERIENCE

MASSACHUSETTS INSTITUTE OF TECHNOLOGY BCS, James DiCarlo Lab

Research Project on Vision Recognition in Macaques

Boston, USA
August 2024 - Present

- Microstimulation and recording data in the anterior Inferior temporal cortex (aIT) in macaques
- Coding processing pipeline of artefact removal for the data collected and analysis of the clean data

OXFORD UNIVERSITY, NDCN, Ben Seymour lab

Research masters project looking at relationship between pain and reward mechanism

Oxford, UK
September 2023 - Present

- Experimental protocol focused on questionnaires and ratings, with supplementary EEG data and physiological measures
- Coding VR in unity for a more realistic and immersive experience for participants.

INSTITUTE OF COGNITIVE NEUROSCIENCE UCL, PATRICK HAGGARD RESEARCH LAB

Internship/Last Year Project

London, United Kingdom
January 2022 – June 2023

- Performed EEG set up and testing to measure readiness potential
- Preprocessed and analysed EEG data using MATLAB toolbox
- Experienced with participant recruitment platform Sona
- Managed Psych toolbox in MATLAB to create stimuli for study

MAX-PLANCK INSTITUTE OF REGENERATIVE MEDICINE GUANGZHOU

Internship

GuangZhou, China
August 2021 – October 2021

- Culturing of Liver stem cells using EGF, PDGF, IGF-1 and utilization of light microscopy to record progress HLSCs regeneration after hepatic injury in vitro
- Performance of Western Blot to identify proteins expressed by each type of embryonic liver stem cell to interpret which is crucial for proliferation
- Identification of mRNA responsible for transcription in HLSCs using microarray analysis
- Obtained mice testing permit to investigate differentiation patterns of embryonic human liver stem cells in immunodeficient mice models in vivo
- Application of previous knowledge on deep learning MATLAB models to accelerate identification of carcinoma cells from microscope images

WORK EXPERIENCE

SIEMENS HEALTHINEERS MAGNETIC RESONANCE SHENZHEN

ShenZhen, China

Internship

June 2021 – August 2021

- Aided fMRI scanning of participants of heart and brain. Training of independent usage of fMRI
- Visualizing whole manufacturing process of a fMRI scan better helped understand physics background of fMRI
- Identification of individual spike artifacts to further isolate and study
- Pre-processing of fMRI data before visualization, including slice time correction and fitting model assumptions using MATLAB

LEADERSHIP

STUDENT REPRESENTATIVE OF BIOMEDICAL SCIENCE (UCL)

January 2021 - December 2022

- Bridge of communication between students and module, course, and department professors
- Recruitment and anonymization of feedback for modules and exam organizations to aid improvement and raise concerns.
- Represent student community on how department finance should be distributed to enhance well-being among students

HUAWEI STUDENT AMBASSADOR (HSD)

London, United Kingdom

Student Ambassador

January 2021 - December 2022

- Organized technology and coding events for students at UCL
- Networking events, research competitions advertisement and career fairs

HARVARD COLLEGE ASSOCIATION LEADERS FOR US-CHINA RELATIONS

HangZhou, China

Teaching Assistant

January 2021 - December 2021

- Organization of two modules: Introduction to Molecular Biology & Machine Learning
- Succeeded in advising high school students and inspiring on future college choices
- Full responsibility on class schedule of students and lecturers for 2 weeks as well as leisure activity
- Increased online lecture participation by introducing special awards based on participation

BIOBOAT

Online platform

Editorial Department Leader

January 2021 - December 2021

- BioBoat Community Leader
- Co-founder of editorial department of BioBoat platform (responsible for intercommunications between laboratories/companies and Biology related major students)
- Created internship information for subscribers and non-professional articles based on research novelties to a wider public.
- Content has spurred a growth of subscribers from last year ranging a large diversity of biology student community all over the world

VOLUNTEERING

RUMANIA DISABLED CHILDREN'S ORPHANAGE (HOUSE OF ANGELS)

Gaesti, Rumania

Volunteer

January 2019 - December 2019

- Organized recruitment of donations for children in orphanage
- Developed social events and intellectual challenges for better group cohesion among children with different disabilities

LANGUAGES AND SKILLS

- Spanish (native) | Chinese(native) | English (C2) | French (Elementary)
- LinkedIn: <https://www.linkedin.com/in/ruohan-liu-b6116a117>
- Coding languages: Matlab, Python, R, SQL
- Useful tools: stata, excel, LaTeX, google collab, EEG setup

RESEARCH PAPERS/POSTER PRESENTATION

- **Representations, consciousness memory in goal-directed behavior (*in preparation*)** With poster on same paper London, UK
- **Exploring the Effort-Reward Paradox in sports under the experimental paradigm of DOMS model** Oxford, UK
Oxford Neuroscience symposium
Inaugural Annual Conference of the Podium Institute for Sports Medicine & Technology 2024

